Министерство образования Республики Беларусь Учреждение образования
«Брестский государственный технический университет»
Кафедра ИИТ

Лабораторная работа №3
По дисциплине: «Обработка изображений в интеллектуальных системах»
По теме: «**Обучение детекторов объектов**»

Выполнил:
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Брест 2026

Цель: осуществлять обучение нейросетевого детектора для решения задачи обнаружения заданных объектов

В-т	Детектор	Датасет
5	YOLO11m	**Счетчики расхода воды:** https://universe.roboflow.com/koer3741-gmail-com/watermeteramrv2/dataset/1

Результаты работы сети:

```
Конфигурация датасета:
train: ../train/images
val: ../valid/images
test: ../test/images
nc: 12
names:
- '0'
- '1'
- '2'
- '3'
- '4'
- '5'
- '6'
- '7'
- '8'
- '9'
- counter
- liters
roboflow:
  workspace: koer3741-gmail-com
  project: watermeteramrv2
  version: 1
  license: CC BY 4.0
  url: https://universe.roboflow.com/koer3741-gmail-com/watermeteramrv2/dataset/1


Количество изображений:
train: 3478
valid: 389
test: 389
```

	from	n	params	module	arguments
0	-1	1	1856	ultralytics.nn.modules.conv.Conv	[3, 64, 3, 2]
1	-1	1	73984	ultralytics.nn.modules.conv.Conv	[64, 128, 3, 2]
2	-1	1	111872	ultralytics.nn.modules.block.C3k2	[128, 256, 1, True, 0.25]
3	-1	1	590336	ultralytics.nn.modules.conv.Conv	[256, 256, 3, 2]
4	-1	1	444928	ultralytics.nn.modules.block.C3k2	[256, 512, 1, True, 0.25]
5	-1	1	2360320	ultralytics.nn.modules.conv.Conv	[512, 512, 3, 2]
6	-1	1	1380352	ultralytics.nn.modules.block.C3k2	[512, 512, 1, True]
7	-1	1	2360320	ultralytics.nn.modules.conv.Conv	[512, 512, 3, 2]
8	-1	1	1380352	ultralytics.nn.modules.block.C3k2	[512, 512, 1, True]
9	-1	1	656896	ultralytics.nn.modules.block.SPPF	[512, 512, 5]
10	-1	1	990976	ultralytics.nn.modules.block.C2PSA	[512, 512, 1]
11	-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
13	-1	1	1642496	ultralytics.nn.modules.block.C3k2	[1024, 512, 1, True]

```
 14                -1  1       0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 15           [-1, 4]  1       0  ultralytics.nn.modules.conv.Concat           [1]
 16                -1  1  542720  ultralytics.nn.modules.block.C3k2             [1024, 256, 1, True]
 17                -1  1  590336  ultralytics.nn.modules.conv.Conv              [256, 256, 3, 2]
 18          [-1, 13]  1       0  ultralytics.nn.modules.conv.Concat           [1]
 19                -1  1 1511424  ultralytics.nn.modules.block.C3k2             [768, 512, 1, True]
 20                -1  1 2360320  ultralytics.nn.modules.conv.Conv              [512, 512, 3, 2]
 21          [-1, 10]  1       0  ultralytics.nn.modules.conv.Concat           [1]
 22                -1  1 1642496  ultralytics.nn.modules.block.C3k2             [1024, 512, 1, True]
 23      [16, 19, 22]  1 1420276  ultralytics.nn.modules.head.Detect            [12, 16, None, [256,
512, 512]]
YOLO11m summary: 231 layers, 20,062,260 parameters, 20,062,244 gradients, 68.4 GFLOPs

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 1/50      2.36G      1.484       1.59      1.224         14        640: 100% ──────────── 870/870 6.8it/s 2:08
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 8.0it/s 6.1s
             all        389       2059       0.57      0.698       0.69      0.298

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 2/50       2.9G      1.465       1.091      1.194         10        640: 100% ──────────── 870/870 6.9it/s 2:06
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 9.1it/s 5.4s
             all        389       2059      0.732      0.765       0.825      0.424

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 3/50       2.9G      1.435      0.9821      1.176          0        640: 100% ──────────── 870/870 6.7it/s 2:10
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 7.9it/s 6.2s
             all        389       2059      0.819      0.796       0.853       0.43

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 4/50       2.9G      1.427      0.9403      1.176         11        640: 100% ──────────── 870/870 6.6it/s 2:11
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 8.3it/s 5.9s
             all        389       2059      0.878      0.863        0.91      0.487

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 5/50      2.91G        1.4      0.8652      1.151         10        640: 100% ──────────── 870/870 6.9it/s 2:06
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 7.9it/s 6.2s
             all        389       2059       0.89      0.878       0.923      0.472

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 6/50      2.91G       1.39      0.8178      1.155          9        640: 100% ──────────── 870/870 7.0it/s 2:05
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 8.5it/s 5.7s
             all        389       2059        0.9      0.891       0.93      0.501

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
45/50      2.91G      1.084     0.4433      1.049         12        640: 100% ──────────── 870/870 7.1it/s 2:03
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 8.1it/s 6.1s
             all        389       2059       0.96      0.969       0.983      0.593

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
46/50      2.91G      1.082      0.442      1.047          7        640: 100% ──────────── 870/870 7.1it/s 2:03
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 7.8it/s 6.3s
             all        389       2059      0.965      0.968        0.98      0.586

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
47/50      2.91G      1.068     0.4335      1.044         12        640: 100% ──────────── 870/870 7.1it/s 2:03
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 9.1it/s 5.4s
             all        389       2059      0.963      0.971        0.98      0.592

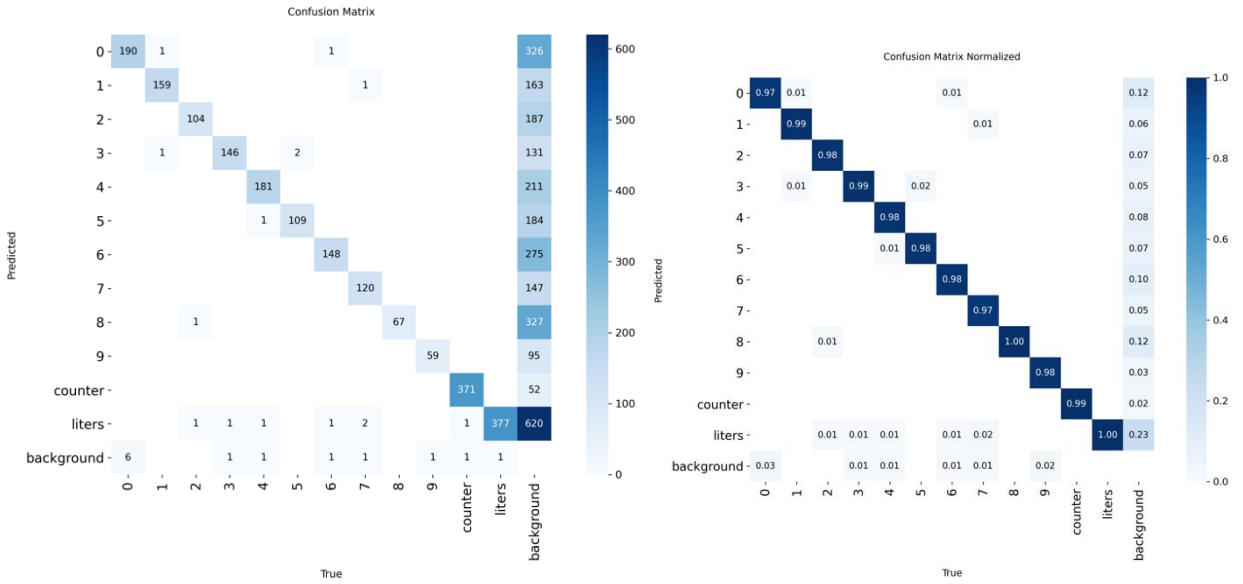
Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
48/50      2.91G      1.065     0.4303      1.038         14        640: 100% ──────────── 870/870 7.0it/s 2:04
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 8.5it/s 5.8s
             all        389       2059      0.961      0.974       0.981      0.592

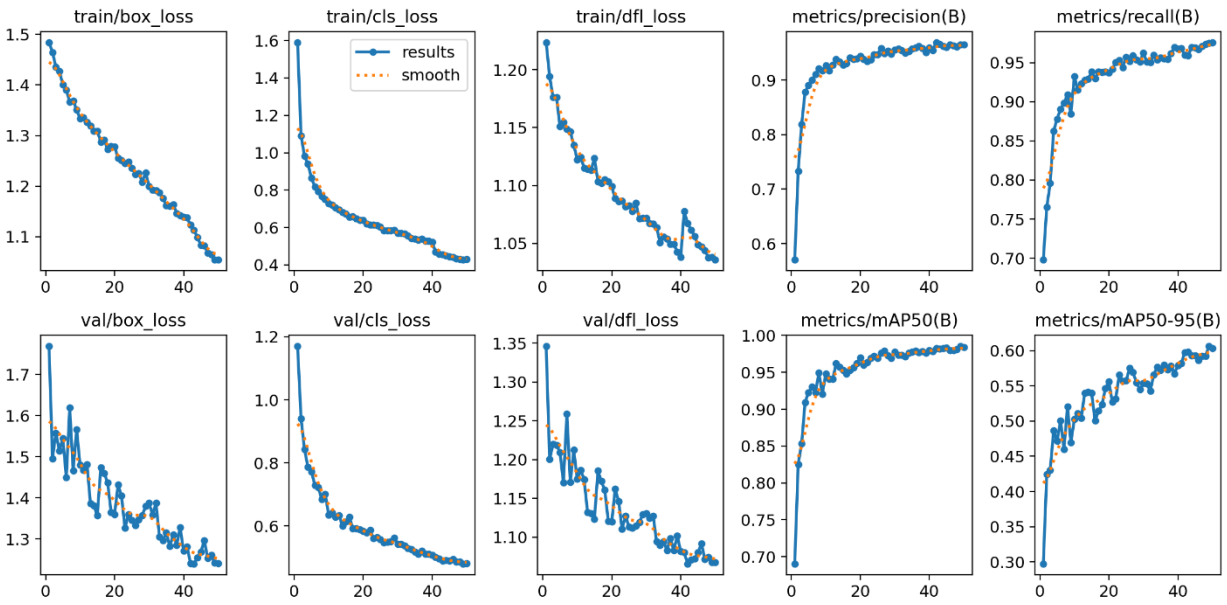
Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
49/50      2.91G      1.055     0.4271      1.038         12        640: 100% ──────────── 870/870 7.0it/s 2:04
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 9.0it/s 5.4s
             all        389       2059      0.965      0.975       0.985      0.606

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
50/50      2.91G      1.054     0.4301      1.036         12        640: 100% ──────────── 870/870 7.0it/s 2:04
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 7.8it/s 6.3s
             all        389       2059      0.965      0.976       0.984      0.603
```

Class	Images	Instances	Box(P	R	mAP50	mAP50-95):
all	389	2105	0.952	0.954	0.969	0.569
0	162	213	0.877	0.939	0.949	0.53
1	108	153	0.901	0.961	0.956	0.522
2	87	90	0.962	0.978	0.964	0.506
3	125	154	0.945	0.901	0.951	0.558
4	132	165	0.966	0.982	0.987	0.597
5	119	167	0.95	0.964	0.972	0.549
6	112	130	0.963	0.931	0.963	0.569
7	100	115	0.98	0.983	0.994	0.565
8	64	75	0.958	0.915	0.965	0.529
9	70	75	0.961	0.933	0.948	0.564
counter	371	383	0.987	0.984	0.992	0.774
liters	377	385	0.976	0.979	0.982	0.567

```
Speed: 1.5ms preprocess, 44.5ms inference, 0.0ms loss, 1.3ms postprocess per image
Results saved to /content/runs/watermeter/test_evaluation
mAP@0.5:      0.9687 (96.87%)
mAP@0.5:0.95: 0.5692 (56.92%)
```





Так как сеть плохо работала на изображениях с интернета было принято решение дообучить ее с более сильной аугментацией

	from	n	params	module	arguments
0	-1	1	1856	ultralytics.nn.modules.conv.Conv	[3, 64, 3, 2]
1	-1	1	73984	ultralytics.nn.modules.conv.Conv	[64, 128, 3, 2]
2	-1	1	111872	ultralytics.nn.modules.block.C3k2	[128, 256, 1, True, 0.25]
3	-1	1	590336	ultralytics.nn.modules.conv.Conv	[256, 256, 3, 2]

```
 4            -1  1    444928  ultralytics.nn.modules.block.C3k2            [256, 512, 1, True, 0.25]
 5            -1  1   2360320  ultralytics.nn.modules.conv.Conv             [512, 512, 3, 2]
 6            -1  1   1380352  ultralytics.nn.modules.block.C3k2            [512, 512, 1, True]
 7            -1  1   2360320  ultralytics.nn.modules.conv.Conv             [512, 512, 3, 2]
 8            -1  1   1380352  ultralytics.nn.modules.block.C3k2            [512, 512, 1, True]
 9            -1  1    656896  ultralytics.nn.modules.block.SPPF           [512, 512, 5]
10            -1  1    990976  ultralytics.nn.modules.block.C2PSA          [512, 512, 1]
11            -1  1         0  torch.nn.modules.upsampling.Upsample        [None, 2, 'nearest']
12       [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat          [1]
13            -1  1   1642496  ultralytics.nn.modules.block.C3k2            [1024, 512, 1, True]
14            -1  1         0  torch.nn.modules.upsampling.Upsample        [None, 2, 'nearest']
15       [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat          [1]
16            -1  1    542720  ultralytics.nn.modules.block.C3k2            [1024, 256, 1, True]
17            -1  1    590336  ultralytics.nn.modules.conv.Conv             [256, 256, 3, 2]
18      [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat          [1]
19            -1  1   1511424  ultralytics.nn.modules.block.C3k2            [768, 512, 1, True]
20            -1  1   2360320  ultralytics.nn.modules.conv.Conv             [512, 512, 3, 2]
21      [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat          [1]
22            -1  1   1642496  ultralytics.nn.modules.block.C3k2            [1024, 512, 1, True]
23      [16, 19, 22]  1   1420276  ultralytics.nn.modules.head.Detect           [12, 16, None, [256,
512, 512]]
YOLO11m summary: 231 layers, 20,062,260 parameters, 20,062,244 gradients, 68.4 GFLOPs
```

```
Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 1/15      5.13G      1.301     0.5686      1.461         18        960: 100% ──────────── 870/870 3.7it/s 3:55
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.2it/s 11.7s
             all        389       2059      0.919      0.924      0.936      0.384

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 2/15      6.31G      1.299     0.5966      1.438         18        960: 100% ──────────── 870/870 3.8it/s 3:50
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.5it/s 10.9s
             all        389       2059      0.912      0.913       0.93       0.41

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 3/15      6.31G      1.299     0.6236      1.448         12        960: 100% ──────────── 870/870 3.8it/s 3:50
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.6it/s 10.6s
             all        389       2059       0.91      0.917      0.916      0.393

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 4/15      6.31G      1.265     0.6008      1.425          7        960: 100% ──────────── 870/870 3.8it/s 3:51
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.5it/s 10.9s
             all        389       2059      0.933      0.938      0.947      0.418

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
 5/15      6.31G      1.279     0.6011       1.43          9        960: 100% ──────────── 870/870 3.8it/s 3:50
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.6it/s 10.7s
             all        389       2059      0.944      0.935      0.953      0.465
Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
10/15      6.31G       1.09     0.4558      1.311         11        960: 100% ──────────── 870/870 3.7it/s 3:55
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.4it/s 11.2s
             all        389       2059      0.949      0.955      0.969      0.519

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
11/15      6.31G      1.076     0.4409      1.299         12        960: 100% ──────────── 870/870 3.7it/s 3:56
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.5it/s 10.9s
             all        389       2059      0.948      0.954      0.971      0.517

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
12/15      6.31G      1.044     0.4277      1.272          9        960: 100% ──────────── 870/870 3.7it/s 3:54
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.4it/s 11.1s
             all        389       2059      0.955      0.961       0.97      0.482

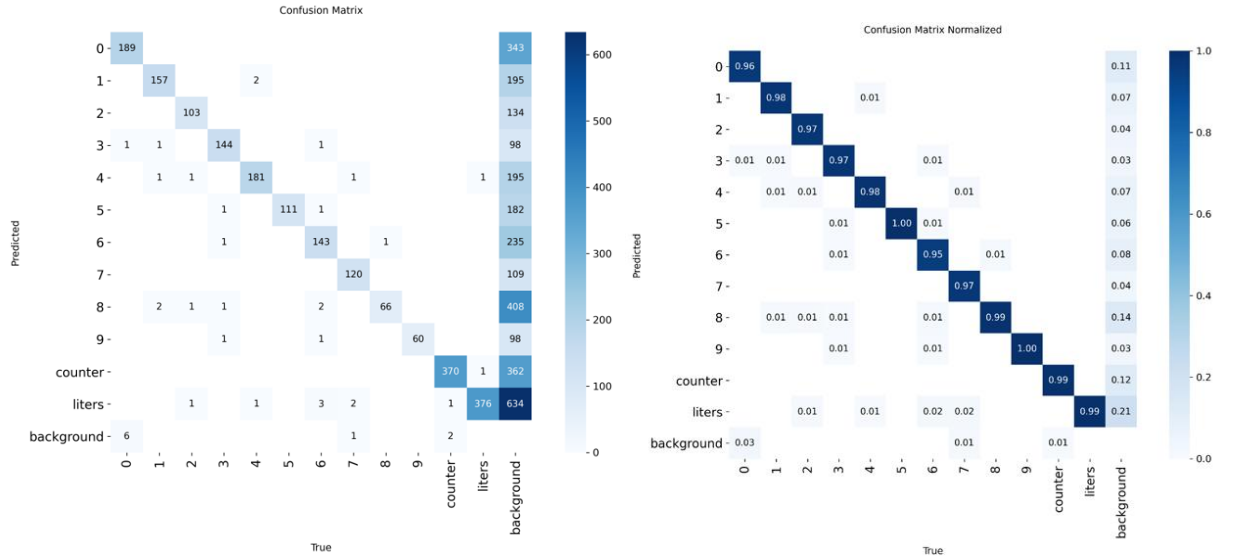
Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
13/15      6.31G      1.027     0.4181      1.262         12        960: 100% ──────────── 870/870 3.7it/s 3:53
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.4it/s 11.2s
             all        389       2059      0.947       0.96      0.976      0.565

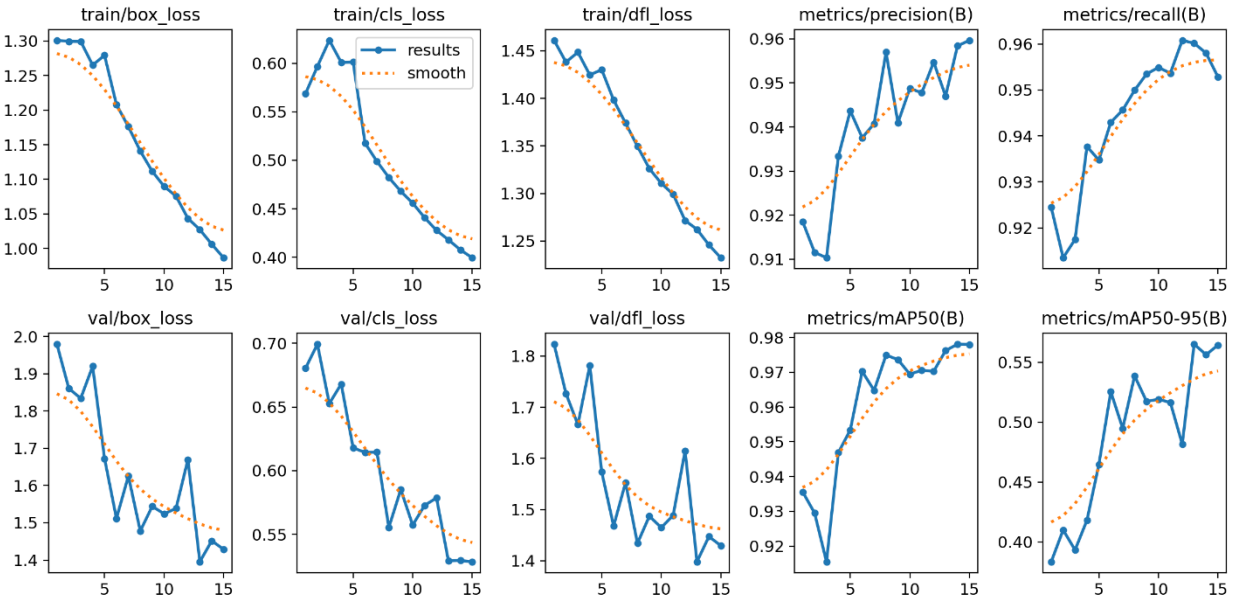
Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
14/15      6.31G      1.006     0.4077      1.246         12        960: 100% ──────────── 870/870 3.7it/s 3:55
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.4it/s 11.1s
             all        389       2059      0.958      0.958      0.978      0.556

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
15/15      6.31G     0.9866     0.3995      1.232         10        960: 100% ──────────── 870/870 3.7it/s 3:55
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ──────────── 49/49 4.6it/s 10.6s
             all        389       2059       0.96      0.953      0.978      0.564
```

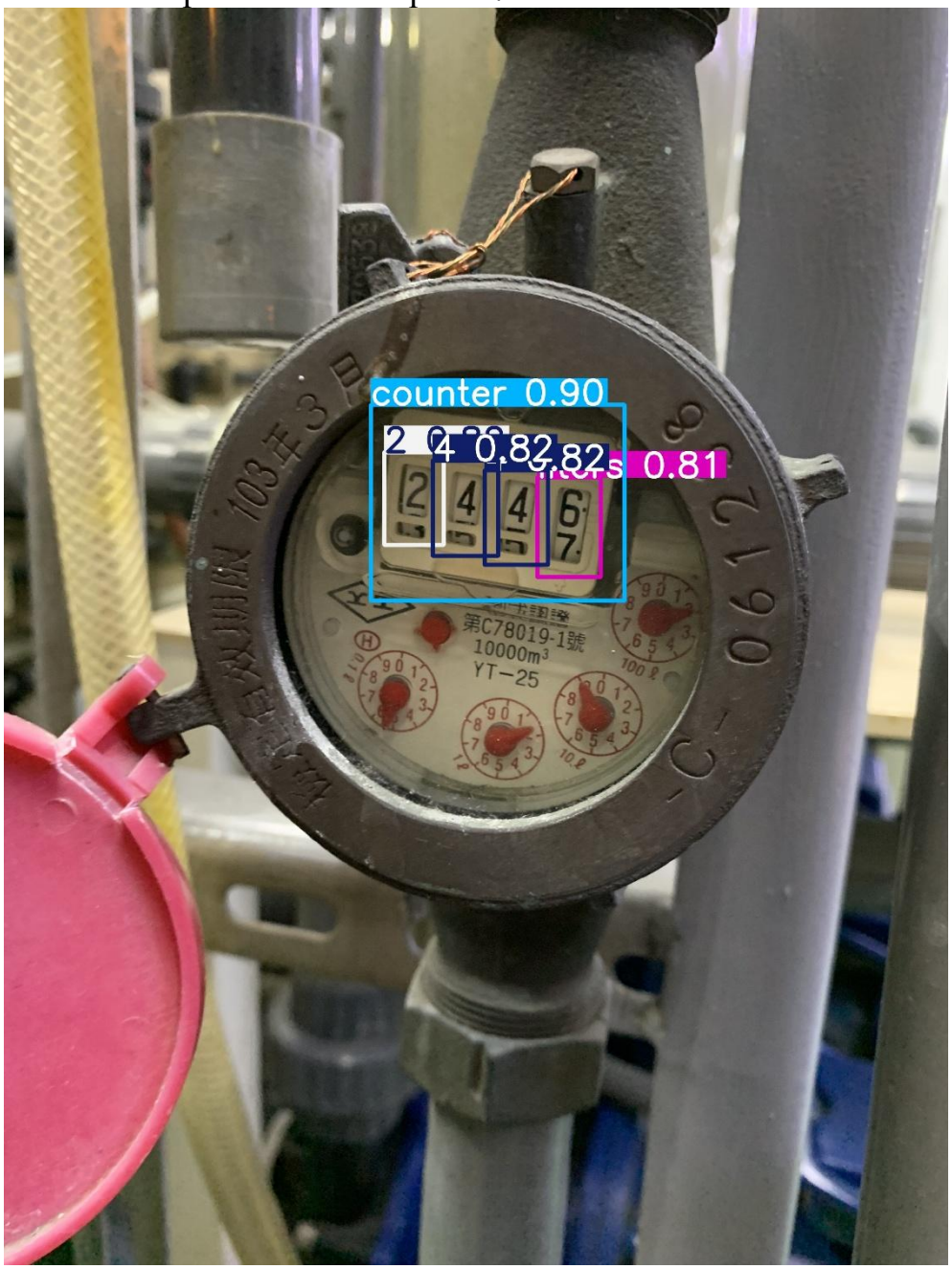
```
YOLO11m summary (fused): 125 layers, 20,039,284 parameters, 0 gradients, 67.8 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
                   all        389       2059      0.946       0.96      0.976      0.564
                     0        151        196      0.891      0.908      0.924      0.494
                     1        122        161      0.922      0.919      0.973      0.542
                     2         97        106      0.968      0.962      0.976      0.574
                     3        124        148      0.975      0.973      0.989      0.586
                     4        148        184      0.975      0.978      0.988      0.567
                     5         96        111      0.916      0.986       0.97      0.527
                     6        125        151       0.97      0.947      0.989      0.598
                     7         98        124      0.985      0.944      0.968      0.569
                     8         60         67      0.874      0.925      0.973      0.546
                     9         58         60      0.939          1      0.986       0.58
               counter        367        373      0.975      0.992      0.993      0.658
                liters        375        378      0.966      0.988      0.988       0.53
Speed: 1.8ms preprocess, 25.9ms inference, 0.0ms loss, 3.0ms postprocess per image
```

График ошибок:





Тестовая картинка из интернета:



Вывод: осуществил обучение нейросетевого детектора для решения задачи обнаружения заданных объектов.